Gold Price Prediction Using Linear Regression

A Quantitative analysis and Machine Learning Pipeline for Market Trend Forecasting

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TEAM 9

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REPO LINK : https://github.com/rahulforcode89/Gold-price-prediction-using-Linear-Regression

Problem and Data

What is the Goal?

Forecast Gold ETF (GLD) closing prices using financial indices and technical indicators via Ordinary Least Squares Linear Regression.

Why use Linear Regression?

Provides clear model explanation, fast statistical validation, and low computational overhead for macroeconomic feature baseline testing.

Dataset Overview

- Rows=1718
- Columns=74
- Shape= (1718,74)

Dataset Source: **https://www.kaggle.com/datasets/sid321axn/gold-price-prediction-dataset KAGGLE**

Model Pipeline and Libraries used

01. Data Ingestion

Load CSV, handle nulls, feature matrix extraction

→

02. Preprocessing

Moving averages, 80-20 train-test split

→

03. Linear regressor

Fit OLS model, optimize weights

→

04. Output & Evaluation

Price forecasts, R^2 score, MAE calculation

Libraries used:

Pandas and NumPy

Data wrangling and matrix math

Scikit Learn

Linear Regression and metrics

Mathplotlib and Seaborn

Visualisations, plots and correlation

Performance metrics and visualisation

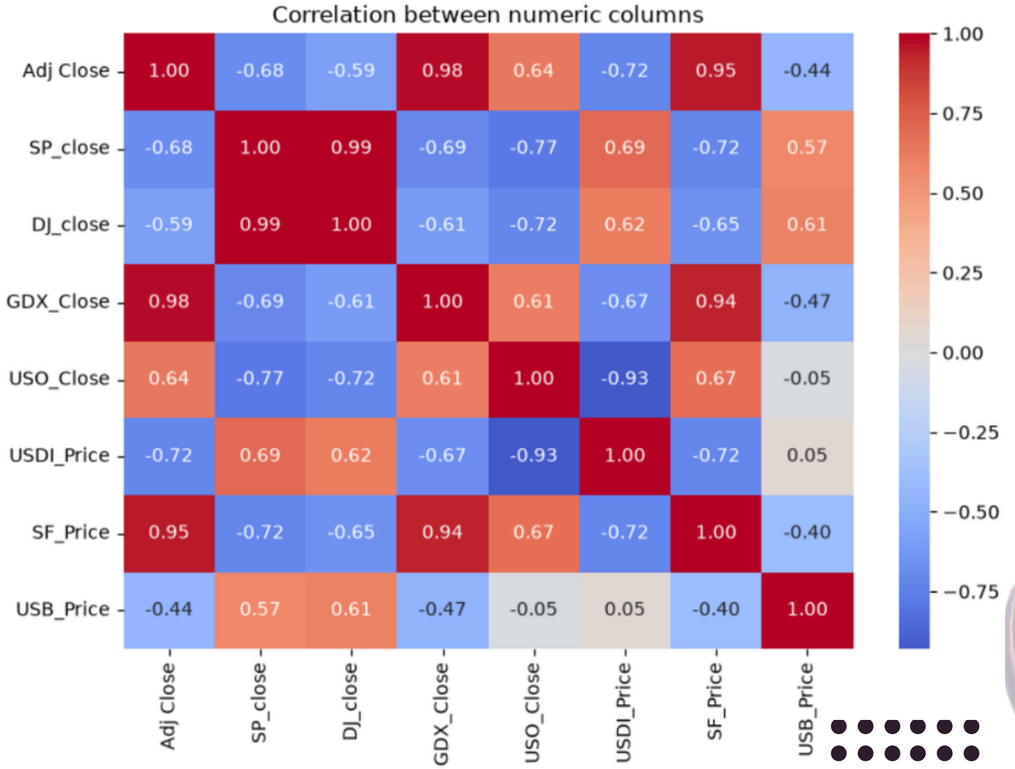
```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print(f"MAE       : {mae:,.3f}")
print(f"MSE       : {mse:,.3f}")
print(f"RMSE      : {rmse:,.3f}")
print(f"R² score : {r2:.3f}")
```

✓ 0.0s

```
MAE       : 3.736
MSE       : 21.663
RMSE      : 4.654
R² score : 0.930
```

DEMO SCREENSHOT



CORRELATION CHART

The Model explains 93% of Test Variation

KEY NUMBERS

- Mean Absolute Error (MAE) = 3.736
- Mean Squared Error (MSE) = 21.663
- Root Mean Squared Error (RMSE) = 4.654
- R2 Score = 0.930

Conclusion

01. What worked and what didn't?

Finding clear correlation between numeric columns, and visualisation of data helped.

Inconsistent data, inability to find linear relation for more complex relations and such weak correlations proved to be hindrances.

02. Who did what?

- Rahul Kumar - Documentation and Preprocessing of data
- Vineet Singh Lodhi - Model tuning and Execution
- Manroj Singh - Building and training model, Slide Deck Design
- Barmavath Harshavardhan Naik - Video Walkthrough Recording

AI DISCLOSURE: Part of the code was written with ChatGPT for efficiency and debugging purposes.
No AI was used to make this Slide Deck except for Slide 2 where Google Gemini was used to phrase the sentences in a more cohesive manner.
Slide Deck made using Canva.

THANK YOU